

# Consolidated claim map — Paper IV v1.22-r3

Columns: claim, Lean declaration, semantic verdict, origin of the verdict.

- **Historical rows (H)** come from `run_v1.2_r1 20_EVIDENCE/E1/E1_RECORD.md` (21 rows). They are inherited by identity: the statement text is unchanged across the `v1.2→r3` diff chain, and the Lean cut is 615/615.
- **New rows (N, run\_v1.22\_r3)** are new. They were produced by a static comparison of the literal frozen headers (`20_EVIDENCE/E1/E1E3_HEADERS_CHECK.json`) and the definitions they use with the EN manuscript text, which is identical to r2 outside the five status paragraphs.
- The new rows **were not part of the original E1**. The Lean types used are not newly elaborated: the headers are source text, and their compilation is the `run_v1.2_r1` build (module records PASS, exit 0, logs hash-bound).

## H — historical rows (run\_v1.2\_r1 E1, inherited by identity)

| ID     | Claim                                         | Lean declaration                                                                                                                                                                                                        | Verdict                                                 |
|--------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| A      | Thm A, all-order additive                     | <code>Erdos81AllOrders.erdos81_all_orders_additive</code><br><code>ExplicitThreshold.erdos81_all_orders_bounded_additive</code>                                                                                         | MATCH                                                   |
| B1     | Thm B (1.1)                                   | <code>Erdos81Unconditional.erdos81_cliquePartition</code>                                                                                                                                                               | MATCH                                                   |
| B2     | Thm B (1.2)                                   | <code>PaperTheorems.erdos81_max_eq</code>                                                                                                                                                                               | MATCH                                                   |
| C      | Thm C                                         | <code>DefectExplicitPublication.rooted_defect_eventual/_maximum</code>                                                                                                                                                  | MATCH                                                   |
| C'a-d  | Thm C' (6.12), (6.13); Cor 6.3 (6.17); (6.18) | <code>fixed_defect_joint_stability_real</code> ; <code>fixed_defect_exact_extremal_edit_real</code> ; <code>E32.cp_classification_of_theoremCPrimeC</code> + <code>FixedExplicit.fixed_defect_stability_explicit</code> | MATCH (X-27 for (6.18))                                 |
| 6.3a   | Prop 6.3a                                     | <code>OptimalTemplateObstruction.*</code>                                                                                                                                                                               | MATCH; X-17 (Lean existential)                          |
| 6.1    | Thm 6.1                                       | <code>IntegralStability...</code> ; <code>chordal_joint_stability_real</code>                                                                                                                                           | MATCH                                                   |
| 6.1a   | Cor 6.1a                                      | <code>chordal_joint_stability_real</code>                                                                                                                                                                               | MATCH; X-18 (Lean real $\tau$ )                         |
| 6.2    | Cor 6.2                                       | <code>ExtremalClassification.chordal_extremal_classification</code>                                                                                                                                                     | MATCH                                                   |
| 6.4    | Prop 6.4                                      | <code>certified*_fractional_bound_all_orders</code> ; <code>uniform*_partition_bound</code>                                                                                                                             | MATCH                                                   |
| 6.5    | Thm 6.5                                       | <code>sequence_arbitrary_orders_*</code>                                                                                                                                                                                | MATCH                                                   |
| 6.6    | Cor 6.6                                       | <code>E35.theoremC_tower*</code>                                                                                                                                                                                        | MATCH                                                   |
| C.4    | (C.4)                                         | <code>E35.NfarE_le_tower_poly</code>                                                                                                                                                                                    | MATCH                                                   |
| 3.1    | Thm 3.1                                       | <code>RC01Final.rc01_uniformRoundingTarget</code>                                                                                                                                                                       | MATCH                                                   |
| D.2    | Prop D.2                                      | <code>DefectSharpPublication.order_three_insufficient</code>                                                                                                                                                            | MATCH                                                   |
| F.3a/b | Lemma F.3a, Cor F.3b                          | <code>exists_clique_additive_defect</code> ; <code>recover_clique_from_edit</code>                                                                                                                                      | MATCH                                                   |
| G.1    | (G.1) annex                                   | <code>BoundedCliqueGap.chordal_gap_linear_cliqueFree</code>                                                                                                                                                             | MATCH                                                   |
| G.2    | Prop G.1                                      | <code>FarExploration.CleanupRigidVerdict.threshold_gt_exp</code>                                                                                                                                                        | historically "not re-extracted"; now covered by row N10 |

## N — new rows (run\_v1.22\_r3, static header comparison)

Common facts, verified for all 13 headers:

- The source sha256 equals both the manifest and the JSON.
- The `theorem` line and the header text match an independent re-extraction.
- The namespace is the one stated.
- The `run_v1.2_r1` module log hash matches, and the module record is PASS with exit 0.
- In `RootPartitionStability`, `SplitMixedGap` and `SplitCompleteSharpValue` the in-scope variables are only  $\{V\}$   $[Fintype\ V]$   $[DecidableEq\ V]$ , plus  $\{G\}$   $[DecidableRel\ G.Adj]$ . There are no hidden hypotheses.

Definitions read:

- $\text{splitBaseline } n \ p = p(n-p) - p(p-1)/2$ , which is  $B_n(p)$ .
- $\text{outsideEdges} = \text{edges with both ends outside } R$ , i.e.  $m = e(G - R)$ .
- $\text{missingIncidences} = \sum_{x \in R} \text{the exterior non-neighbours of } x$ , i.e.  $A$ .
- $\text{graphFamDistNorm } F \dots n^2 = \text{min edit distance to the family divided by } n^2$ , i.e.  $d(G)$ .
- $\text{allSplitSupports} = \text{all } S_C = K_{CVI}\{V \setminus C\} \text{ with } C \subseteq V$ .
- $\text{CertifiedFractionalOptimum } G \ w = \text{a primal and a dual of equal value } w$ ; therefore  $w = W^*(G)$  and  $e - w = F_4(G)$ .
- $\text{NearH1Calibration.eps} = 10^{-12}$ ,  $\text{eta} = 10^{-16}$ .
- **The chordality predicates `PaperIV.IsChordal`, `FarRounding.IsChordal` and `SimpleGraph.IsChordal` are equivalent by `Iff.rfl (ChordalBridge)`.**

| ID | Claim (EN)                                                                            | Declaration                                                                          | Hypotheses and conclusion compared                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Verdict                                                                     |
|----|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| N1 | Thm 5.0, strengthened window (§5, (5.0))                                              | <code>PaperIV.NearH1LocalConstructor.exists_near_partition_paid_by_root_sharp</code> | Text: chordal, $n \geq 4 \cdot 10^{12}$ , $d(G) < \varepsilon_0$ , $F_4 \geq n^2/6 - \eta_0 n^2 \Rightarrow$ a clique $R$ with $ 3p - n  \leq n/50$ and $Q$ of order $\leq 4$ with $ Q  \leq B_n(p) - m/20 - A/2$ . Lean: identical hypotheses ( <code>hdist</code> with scale $n^2$ , <code>hnear</code> as $e - w$ ), identical conclusion. It is a local theorem; no global localization is included.                                                                                                                                                                                                                                                                              | <b>MATCH</b>                                                                |
| N2 | Thm 5.0, earlier interface ("the earlier window $n/4 \leq p \leq n/2$ remains valid") | <code>...exists_near_partition_paid_by_root</code>                                   | Same hypotheses and conclusion with the window $n/4 \leq  R  \leq n/2$ . The text derives $ 3p - n  \leq n/50$ from the ratios and attributes the stronger window to N1, not to this declaration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>MATCH</b>                                                                |
| N3 | Cor 5.4 (5.17), root geometry                                                         | <code>PaperIV.NearCriticalDichotomy.chordal_far_or_criticalRoot</code>               | $\exists N \forall n \geq N \forall \text{chordal } G \forall \text{certified } w: F_4 < n^2/6 - \eta_0 n^2$ (strict, as in the text) $\vee \exists R, C$ with $R$ a clique of $G$ , $ 3 R - n  \leq n/50$ , $C$ nonempty, $d_E(G, S_C) \leq \varepsilon_0 n^2$ , $ 3 C  - n  \leq n/10^4$ , and $  R  -  C   \leq n/100$ . <b>C is not asserted to be a clique of <math>G</math></b> (the text calls it a "comparator core"; correct). The last conjunct is a difference of cardinalities, <b>not a symmetric difference</b> ; the text does not assert it, so Lean is slightly stronger.                                                                                            | <b>MATCH</b> (Lean stronger)                                                |
| N4 | Cor 5.4 (5.18), density                                                               | <code>...chordal_far_or_edge_density</code>                                          | Same regime and the same first disjunct $\vee  e(G) - 5n^2/18  \leq n^2/10^4$ . The text states (5.17) and (5.18) "in the second branch". Since both theorems share the literal first disjunct, taking $\max(N, N')$ yields both second-branch conclusions together. The text disclaims an exact density identity ("5/9 describes the asymptotic density").                                                                                                                                                                                                                                                                                                                           | <b>MATCH</b> (with max of thresholds, which the text implies)               |
| N5 | Dichotomy (1.5)                                                                       | <code>PaperIV.HybridDichotomy.chordal_far_or_nearStructure</code>                    | $\text{far} \vee \text{Nonempty } (\text{NearStructureWitness } G) \cdot \text{The structure fields are: } \text{core} \text{ (nonempty) with the edit bound to } S_{\text{core}} \text{ (the complete-split comparator); } \text{regularized} : \text{RegularizedRoot } G, \text{ whose } \text{isClique} \text{ field makes the root a clique of } G; \text{ packing with } \text{IsK34Packing } G \text{ (a packing of the original graph); and } \text{accounts} \text{ with the physical ledger and the improved } 117/1825, 12687/20000 \text{ bound. The text says "the alternatives are exhaustive; ... not asserted to be incompatible", and the disjunction is inclusive.}$ | <b>MATCH</b>                                                                |
| N6 | Optimal quadratic coefficient (6.5a), necessity                                       | <code>PaperIV.SharpConstantOptimality.erdos81_quadratic_constant_optimal</code>      | For rational $c, C$ and a chordal bound by unrestricted clique partitions ( <code>CliquePartition</code> , no order bound; this matches "even when cliques of arbitrary order are allowed"), $1/6 \leq c$ . The real extension in the text is valid: given real $c < 1/6$ and $C$ , choose rational $c' \in (c, 1/6)$ and $C' \geq C$ . Then $cn^2 + Cn \leq c'n^2 + C'n$ for $n \geq 0$ , which contradicts the rational theorem. The header does not quantify over reals, and the text says so.                                                                                                                                                                                     | <b>MATCH</b> (real case by rational enlargement, a valid one-line argument) |
| N7 | Optimal quadratic coefficient, least element                                          | <code>...erdos81_quadratic_constant_isLeast</code>                                   | $\text{IsLeast } \{c \in \mathbb{Q} \mid \exists C N, \text{ bound}\} (1/6)$ . Membership requires Theorem A ( $M(n) + b \leq n^2/6 + (1/6 + b)n$ for $n \geq 1$ ); the text says "combines it with Theorem A". The set is over rationals.                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>MATCH</b> (rational set)                                                 |
| N8 | Optimal linear coefficient (6.5)                                                      | <code>PaperIV.LinearCoefficient.linear_coefficient_optimal</code>                    | For rational $c < 1/6$ and rational $B$ : $\exists N \forall n \geq N \exists \text{chordal } G$ with every clique partition $Q$ satisfying $ Q  > n^2/6 + cn + B$ . The real extension in the text (rational $c' \in (c, 1/6)$ , $B' \geq B_0$ ) is valid. The header is <b>existential in <math>G</math></b> ; the text names the complete-split witness and writes $\text{cp}(G) = M(n)$ , which comes from (6.2)/Thm B, not from this header. This is the same nature as X-17. The additive constant is not minimized (the text says so).                                                                                                                                         | <b>MATCH</b> : observation of X-17 type (existential witness)               |
| N9 | Identity (6.7b)                                                                       | <code>PaperIV.RootPartitionStability.sum_rootPieceDefect_eq</code>                   | Finite $V, R$ a clique, any clique partition $Q$ (no order bound): $\Sigma \text{rootPieceDefect } R \ K =  Q  - B_n( R ) + A + 3m$ .<br>$\text{rootPieceDefect } R \ K = \text{numericDefect } ( K \cap R ,  K \setminus R ) = 1 + C(a, 2) + 3C(b, 2) - ab$ , which is $d_R(K)$ of (6.7a). It uses the actual root and unrestricted pieces, not the capped mixed gain.                                                                                                                                                                                                                                                                                                               | <b>MATCH</b>                                                                |

|     |                                           |                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                 |
|-----|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| N10 | Prop D.3 (D.6)                            | PaperIV.SplitMixedGap.<br>mixed_gap_zero <b>and</b> PaperIV.<br>SplitCompleteSharpValue.<br>exists_sharp_<br>cliquePartition_allParities | $\text{Core} \cap \text{Hosts} = \emptyset$ , $2 \leq k \leq h$ . The first gives: every fractional packing has value $\leq 2C(k,2)$ , and some clique partition has $\Sigma \text{gainOf} = 2C(k,2)$ . The first header alone <b>does not bound the piece order</b> , so its partition might contain pieces of order $\geq 5$ and need not be a mixed packing. The second gives a partition $Q'$ of order $\leq 4$ with $ Q'  = kh - C(k,2)$ (also minimal). The edge-count identity (1.3a), $g(Q') +  Q'  = e(S) = C(k,2) + kh$ , gives $g(Q') = 2C(k,2)$ with $K_2/K_3/K_4$ pieces, i.e. an integral mixed packing of gain $2C(k,2)$ . Hence $W(S) = W^*(S) = 2C(k,2)$ . The text cites both declarations ("together with the bounded construction"). Equality of optimal values is not claimed as integrality of the polytope (the text says "it does not assert that every vertex ... is integral"). | <b>MATCH</b> (needs both declarations plus (1.3a), all present) |
| N11 | Prop G.1 (G.2)                            | FarExploration.<br>CleanupRigidVerdict.<br>threshold_gt_exp                                                                              | CleanupAtWith $\gamma \in (\varepsilon/30) \cup (\varepsilon/4)$ $N_0$ , unfolded: $\forall n \geq N_0 \forall G \forall$ fractional $x$ with triangle mass $\geq (\varepsilon/30)n^2 - 1$ , $\exists \gamma$ with weighted codegree $\leq \gamma$ , triangle mass $\geq C$ , and gain loss $\leq (\varepsilon/4)n^2$ . This is literally the contract in the text. With rational $\varepsilon > 0$ , real $\gamma \leq 1/2$ , real $t \geq 4$ and $7\varepsilon \leq e^{-t} \Rightarrow e^{t^2/16} < N_0$ . The text notes "the precision parameter is rational in the Lean declaration". It is a lower bound for this contract only, as the text states.                                                                                                                                                                                                                                                | <b>MATCH</b> (closes historical G.2 "not re-extracted")         |
| N12 | Prop G.1, "in particular $N_0 > e^{72}$ " | ...threshold_gt_exp_seventy_two                                                                                                          | $\varepsilon \leq 1/(7e^{34}) \Rightarrow e^{72} < N_0$ ( $t = 34$ , $t^2/16 = 72.25 > 72$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>MATCH</b>                                                    |

The 13 headers map to the 12 N-rows; N10 uses two declarations.

**Supplementary E1/E3 verdict:** 12/12 new rows MATCH. One new observation, of X-17 type, is recorded under N8 (existential witness in the linear-coefficient header;  $\text{cp}(G) = M(n)$  comes from other declarations). No translation in `SEMANTIC_COVERAGE_SUPPLEMENT.md` was found to be unjustified. Its four warnings — C is not a clique of G; a size difference is not a symmetric difference; D.3 also needs the bounded constructor; real coefficients go through rational enlargement — are all correct.